A New Global Database of Seismic Vulnerability Models for Improved Regional Risk Assessments

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ABSTRACT

This study introduces a comprehensive global vulnerability database designed to enhance regional seismic risk assessments by addressing the limitations of existing databases, which often relied on oversimplified models that fail to capture storey-level engineering demand parameters and some seismic response patterns. Covering more than 1,400 building classes, the database incorporates an advanced numerical modelling approach, utilising stick-and-mass multi-degree-of-freedom oscillators. Key contributions include: 1) a suite of updated vulnerability functions for structural damage across building types; 2) a set of region-specific storey-loss functions, which use storey demands like drifts and floor accelerations to produce more robust estimates of losses due to non-structural damage; and 3) a dataset of vulnerability functions for losses in building contents. To demonstrate the value of the new database, we employ the new vulnerability models in global risk analysis to calculate a number of risk metrics fundamental for earthquake risk mitigation.